

Machine learning with artificial neural networks

Workshop MLSpeciationGenomics 2025

Christoph Mayer and Petar Arsic

June 2025

1 Introduction

This PDF contains recommended introductory and supplementary materials for the Machine learning lectures of the MLSpeciationGenomics workshop.

1.1 Recommended introduction before the workshop

As a first introduction to neural networks we recommend four videos by 3blue1brown. The content gets increasingly difficult to follow, but you should get an idea how it works. Please ask questions in the lecture!

- But what is a neural network? — Chapter 1, Deep learning
- Gradient descent, how neural networks learn — Chapter 2, Deep learning
We will not need a detailed knowledge about the backpropagation algorithm for the workshop, so you skip these videos if you want.
- What is backpropagation really doing? — Chapter 3, Deep learning (optional)
- Backpropagation calculus — Chapter 4, Deep learning (optional)

Furthermore, we recommend the following video, which provides a short, but nice overview for fully connected and convolutional neural networks.

Neural Network Architectures & Deep Learning

Convolutional neural networks are of fundamental importance not only for identifying patterns in images, but also for identifying patterns in molecular sequences.

As an introduction to convolutional neural networks, we recommend the first part of this video by 3blue1brown. The second part, which start with the introduction of the fft algorithm will not be important for the workshop.

The next video we recommend is long, but covers everything until image segmentation.

Maybe the presentation which leaves the fewest open questions in view of how convolutional layers work is this video: In the first part, it shows in detail how an image is

processed that has three input channels. The mathematical explanations in the second part can be skipped if you want.

Convolutional neural networks are also explained in the following two online papers: (i) Convolutional Neural Networks, Explained and (ii) Convolutional Neural Networks In Depth by Rajesh Katta.

1.2 Installing the necessary software

For the workshop you will need a system with Python installed as well as a working installation of the Tensorflow 2 and sklearn libraries. It is recommended to install the libraries in a virtual environment, e.g. using Miniforge/Conda.

If you are not able to install Python on your computer, you can also use Python on some online System such a Google-Colab.

1.3 Hands-On Introduction to TensorFlow

If you now want to start implementing your own neural networks, we recommend the video series by Aladdin Persson. This is a series of 20 medium length videos that explain how to install and use the Tensorflow 2 library to construct neural network classifiers. All code for his lecture series is available here: Code for the video tutorials, Alladin Persson

In the first video, the installation and setup of Tensorflow is described. You can skip this video if you already have this installed.

TensorFlow Tutorial 1 - Installation and Setup Deep Learning Environment (Anaconda and PyCharm)

In the second video Tenforflow tensors are described. This can be skipped if you want. The introduction lacks mathematical background, and only give you an brief idea what a tensor is in Tensorflow. For our purposes, tensors are scalars, vectors and multi dimensional matrices. Actually, it is not essential for us to know much more at the moment, since the tensors are processed for us in the Tensorflow library. If you are interested you can watch the video anyway.

TensorFlow Tutorial 2 - Tensor Basics

In the third video, a simple neural network is introduced for the classification of hand written numbers: TensorFlow Tutorial 3 - Neural Networks with Sequential and Functional API.

In this video, normal fully connected layers have been used to classify images. Dense layers do not take into account distances between points, i.e. neurons corresponding to neighbouring pixels do not need to be close to one another in the input layer. To take into account spatial distances we need convolutionary layers.

TensorFlow Tutorial 4 - Convolutional Neural Networks with Sequential and Functional API

Please implement the code shown in this lecture.

TensorFlow Tutorial 5 - Adding Regularization with L2 and Dropout

Please implement the code shown in this lecture.

If you want you can watch tutorials 6-9, but they are beyond the scope of the workshop. If you want to use Tensorflow yourself after the workshop, we recommend to watch the lectures 10, 11, 12, 13, 17, 20.

1.4 Detailed introduction, beyond the workshop

The field of neural networks and deep learning is wide. If you are interested in a learning this in detail we recommend first the famous video lecture series by Andrew Ng. It consists of a large number of relatively short videos. The first series of 43 videos gives an introduction to neural networks in general:

Neural network introduction by Andrew Ng

If you are interested in a detailed introduction of convolutional neural networks, we recommend the video series by Andrew Ng with 42 short videos on computer vision and convolutionary networks. We recommend watching videos 1-12. If you think you want to specialise in this direction, you can watch more. Course C4W1L01 Computer Vision

Another video series about programming in Python, data science and machine learning we recommend is the one by "DigitalSreeni". He produced more than 360 medium length videos in these fields: Video tutorials by DigitalSreeni.

We recommend these videos to learn more about special topics you are interested in, e.g. image segmentation.